

```
%nprocshared=64
%mem=320GB
%chk=PbRhaC8C8.chk
# opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc
```

PbRhaC8C8

0 1

O	2.21561400	0.03500400	-0.68434100
O	1.40310600	-3.36138000	-1.67325500
O	-1.61624400	-3.42435800	-0.26047200
O	-0.20661300	1.46248700	-0.90819800
C	2.40635000	4.14161300	-0.51098600
O	3.21841000	3.37918000	0.37251400
C	3.37704700	2.03606500	-0.04025600
C	2.03618800	1.30855700	-0.09592300
C	1.07004300	2.08183300	-0.95769300
C	1.00283500	3.53403700	-0.49962700
O	-2.07480100	-2.62535000	-2.27769200
O	1.92378100	-2.57725700	0.32978900
C	2.17280400	-3.31127200	-0.66164500
C	3.41551300	-4.16894100	-0.69373200
C	4.57906500	-3.62112000	0.10411600
O	4.90035300	-2.31171600	-0.44412700
C	5.79402800	-4.53650200	0.01299900
C	7.00549100	-4.06949500	0.80970900
C	8.17448300	-5.04059600	0.72075200
C	9.39703200	-4.57243700	1.50012300
C	10.56281100	-5.54851900	1.41144400
C	5.18061500	-1.32145700	0.41446400
O	5.08483400	-1.40982200	1.61620700
C	5.65181000	-0.08573700	-0.31563200
C	5.55032400	1.15528700	0.55915000
O	4.17169400	1.39233400	0.88917600
C	2.43058900	5.57407500	-0.03637500
O	0.04682700	4.24837300	-1.26209300
C	-5.00549300	1.15191300	3.33992500
C	-5.37467400	0.79383200	1.87245800
C	-4.36256400	1.41604000	0.92272400
O	-4.09720900	2.73468200	1.34186200
C	-3.26701900	2.76069500	2.51314200
C	-3.54044800	1.55817900	3.44207900
C	-1.80225700	2.88314400	2.13735300
O	-3.26360300	1.90519500	4.78928200
O	-4.87020300	1.41034100	-0.36647900
O	-5.30884700	0.07633600	4.19782600
O	-5.43782600	-0.61406600	1.73042600
C	-3.90842200	1.23293900	-1.41625400
C	-4.34107800	0.01330400	-2.20493500
C	-3.80182800	2.45772400	-2.31247500

C	-4.34438600	-1.24245200	-1.38238800
O	-4.89132900	-2.24744800	-2.04993000
O	-3.89706000	-1.33219900	-0.25262500
C	-4.93151800	-3.57837200	-1.46632400
C	-3.44338300	3.74398500	-1.58366400
C	-3.17249000	4.90159300	-2.53397100
C	-2.89390000	6.21096400	-1.80568000
C	-2.56135900	7.35473300	-2.75506000
C	-3.55878800	-4.28572000	-1.42715500
C	-2.35363200	-3.36574400	-1.30081800
C	-5.65076500	-3.62204200	-0.11967200
C	-6.93669300	-2.81382500	-0.06871300
C	-7.62988900	-2.92170500	1.28208200
C	-8.88172000	-2.06032200	1.38254200
C	-9.54600800	-2.14839600	2.75061700
C	6.27698200	2.36321000	-0.01827100
C	6.38576800	3.52204900	0.96459200
C	6.95431600	4.78047900	0.32558500
C	7.05476100	5.94835100	1.29789100
C	7.60446900	7.21212900	0.64895200
H	2.82029000	4.07558000	-1.53092000
H	3.81961700	2.01340400	-1.04716100
H	1.63577400	1.22948200	0.92232600
H	1.43986800	2.05957800	-1.99560300
H	0.63289500	3.56806100	0.52974300
H	3.13422600	-5.13535600	-0.25965200
H	3.69204200	-4.36009500	-1.73283300
H	4.28158900	-3.47766300	1.14331000
H	5.48512400	-5.52648300	0.36905700
H	6.06575000	-4.65962800	-1.04266500
H	7.33222700	-3.08843600	0.44766400
H	6.72158200	-3.92888200	1.85871700
H	7.86159100	-6.02552000	1.09209500
H	8.44890400	-5.18856000	-0.33235900
H	9.70741500	-3.58873900	1.12617100
H	9.11841400	-4.42196300	2.55064900
H	10.28456200	-6.53002300	1.80799300
H	10.87765700	-5.69145500	0.37297100
H	6.69827800	-0.26374400	-0.58671300
H	5.09986200	0.02513700	-1.25309700
H	1.80842300	6.19817300	-0.67967300
H	3.45127100	5.96148800	-0.05426200
H	2.04779300	5.64092000	0.98544700
H	0.34707600	4.29846400	-2.17751300
H	-5.61019800	2.00743600	3.65840900
H	-6.36345400	1.20419100	1.64270200
H	-3.43212400	0.83046300	0.94515400
H	-3.58307800	3.65204900	3.06274100
H	-2.91834700	0.70004900	3.15279700
H	-1.18055700	2.96427600	3.03405200

H	-1.64273200	3.77805700	1.53275800
H	-1.45927800	2.01444500	1.56929600
H	-2.31586400	1.84316600	4.93790000
H	-5.31902700	-0.70378700	3.62349300
H	-4.80855900	-0.90705100	1.05038800
H	-2.93822900	1.03365900	-0.94750300
H	-5.34770600	0.16005300	-2.60739200
H	-3.68307000	-0.15632100	-3.06395200
H	-3.04862100	2.23509800	-3.07859700
H	-4.75114300	2.59066100	-2.84587500
H	-4.25084200	4.00569700	-0.89540700
H	-2.56281900	3.59380300	-0.94781400
H	-2.31741300	4.65280900	-3.17711200
H	-4.02839800	5.03552600	-3.20949100
H	-3.76687800	6.47672700	-1.19605400
H	-2.06592700	6.05690500	-1.10518000
H	-1.67044000	7.12510300	-3.34834400
H	-3.38252600	7.53880200	-3.45522800
H	-3.56672600	-4.97661700	-0.58422800
H	-3.40820000	-4.86054300	-2.34309400
H	-5.86799500	-4.67944700	0.06804200
H	-4.97521400	-3.29492100	0.67482400
H	-6.71760600	-1.75931000	-0.26801800
H	-7.61485800	-3.14174400	-0.86770700
H	-7.88825700	-3.97004300	1.48253300
H	-6.92615400	-2.61825800	2.06592700
H	-8.61431200	-1.01843700	1.16779400
H	-9.59429600	-2.35744800	0.60266200
H	-9.84491100	-3.17656600	2.97783800
H	-8.86008200	-1.82416400	3.53955400
H	7.27927700	2.03540900	-0.31611600
H	5.79137400	2.69732100	-0.94236300
H	5.39522700	3.74350000	1.36892300
H	7.01366600	3.21937200	1.81248600
H	7.94588400	4.57009500	-0.09760200
H	6.31984700	5.07150100	-0.52269500
H	6.06176300	6.14980000	1.71910400
H	7.68947700	5.65981800	2.14509200
H	8.60793500	7.04300000	0.24598000
H	6.96828000	7.53626900	-0.18071700
H	-0.84461100	2.15417000	-1.12861400
H	2.49172800	-0.61203400	-0.01764400
Pb	-0.10963300	-1.70487300	-0.98345900
H	5.99224700	0.92009900	1.53105600
H	-5.55467500	-4.09002100	-2.20118200
H	-2.36920300	8.28703700	-2.21670500
H	7.66799000	8.03947800	1.36114100
H	11.43059000	-5.19640900	1.97590800
H	-10.44043800	-1.52199400	2.81035000

C H O 0

6-311+g(2d,p)

Pb 0

S 4 1.00

591.61124370 0.22126521076D-03

46.757232559 0.56961959130D-02

20.746462696 -0.21374063831

14.610796419 0.40502620616

S 2 1.00

20.222637612 -0.83541883299D-01

6.4767324865 0.97910892388

S 1 1.00

1.6600600927 1.0000000

S 1 1.00

0.80431655001 1.0000000

S 1 1.00

0.22627039020 1.0000000

S 1 1.00

0.84014530665D-01 1.0000000

P 3 1.00

15.189102118 0.61952303583

14.693144415 -0.72498497086

6.8705890048 0.37680007984

P 3 1.00

2.2021426123 0.40196284806

1.2209125119 0.46058131862

0.63367559815 0.19367655397

P 1 1.00

0.28202837058 1.0000000

P 1 1.00

0.11333375666 1.0000000

P 1 1.00

0.43948707430D-01 1.0000000

D 6 1.00

61.315369628 0.33870800787D-03

12.372195840 0.13788683942D-01

6.9254944983 -0.75979608103D-01

2.3319539939 0.28113784298

1.2108730003 0.44474512269

0.60090478506 0.35326874351

D 1 1.00

0.28135869813 1.0000000

D 1 1.00

0.11500000000 1.0000000

F 1 1.00

1.0000000 1.0000000

F 1 1.00

0.2896200 1.0000000

PB	0	
PB-ECP	3	60
f potential		
2		
2	3.88751200	12.20989200
2	3.81196300	16.19029100
s-f potential		
4		
2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
p-f potential		
6		
2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
d-f potential		
6		
2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600
2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100